



# LOAN PORTFOLIO & DEFAULT ANALYTICS

Interactive Power BI portfolio project documenting dashboard design, DAX measures, risk analysis, and Power BI Service deployment.

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Technology: Power BI Desktop, Power BI Service, Power Query, DAX, SQL Server, GitHub

Project documentation prepared for portfolio / GitHub use

# 1. Executive Summary

Loan Portfolio & Default Analytics is a three-page Power BI report designed to convert loan application data into an interactive business-intelligence experience. The report combines executive KPIs, applicant profiling, portfolio segmentation, default analysis, Key Influencers, Decomposition Tree exploration, slicers, bookmarks, drill-through concepts, and dynamic risk insights.

The final visual system uses a dark financial theme with navy surfaces, cyan primary accents, purple secondary accents, orange warnings, and red risk indicators. The design was intentionally standardized across all three pages so the report reads as one product rather than three unrelated dashboards.

The report was published to Power BI Service under the final report name Loan Portfolio & Default Analytics. The PBIX file is stored in the project GitHub repository for portfolio sharing.

## Key portfolio metrics captured in the report

| Metric                    | Current value shown in dashboard |
|---------------------------|----------------------------------|
| Total Loan Amount         | ₹33bn                            |
| Average Loan Amount       | ₹127.58K                         |
| Overall Default Rate      | 11.61%                           |
| Average Interest Rate     | 13.49%                           |
| Defaulted Loans           | 29.65K                           |
| Low Credit Exposure       | ₹14.9bn                          |
| Low Credit Exposure Share | 45.7%                            |
| Low Credit Default Rate   | 12.58%                           |

Note: dashboard values are presented as shown in the completed Power BI report and may change when filters or model context change.

## 2. Dataset & Analytical Scope

The underlying loan table is named Loan+Dataset+Link. The model contains applicant, financial, credit, loan, employment, demographic, and default-related fields.

Core fields used in the report include: Age, Age Group, CreditScore, Credit score Bins, Default, DTIRatio, Education, EmploymentType, Income, Income Bracket, InterestRate, LoanAmount, LoanID, LoanPurpose, LoanTerm, MaritalStatus, MonthsEmployed, NumCreditLines, HasMortgage, HasDependents, HasCoSigner, and Year/date fields.

The dashboard is primarily descriptive and diagnostic. It summarizes what is happening in the portfolio, compares applicant segments, and surfaces factors associated with default. It does not claim that any single demographic or financial variable causes default.

The report architecture at publication time uses SQL Server as the upstream source, a Power BI Dataflow / semantic-model layer, and Power BI Service for report delivery. The current workspace has a Dataflow refresh limitation for linked/computed/incremental-refresh features that require Premium capacity; therefore scheduled refresh is intentionally not enabled for this portfolio deployment.

### Data preparation and modeling goals

1. Validate field types and date handling.
2. Create categorical bands for analysis.
3. Build reusable DAX measures rather than hard-coded visual calculations.
4. Provide consistent filter context across pages.
5. Design visuals around business questions rather than chart variety alone.

### Calculated category fields

```
Credit score Bins = IF( 'Loan+Dataset+Link'[CreditScore] <= 450, "Very Low",  
IF('Loan+Dataset+Link'[CreditScore] <= 550, "Low", IF('Loan+Dataset+Link'[CreditScore] <= 650, "Medium",  
"High") ) )
```

```
Age Group = IF('Loan+Dataset+Link'[Age] <= 19, "Teen", IF('Loan+Dataset+Link'[Age] <= 39, "Adults",  
IF('Loan+Dataset+Link'[Age] <= 59, "Middle Age Adults", "Senior Citizens") ) )
```

```
Income Bracket = SWITCH( TRUE(), 'Loan+Dataset+Link'[Income] < 30000, "Low Income",  
'Loan+Dataset+Link'[Income] < 60000, "Medium Income", "High Income" )
```

```
Year = YEAR('Loan+Dataset+Link'[Loan_Date_DD_MM_YYYY])
```

### 3. Dashboard Page 1 - Loan Portfolio & Default Overview

Purpose: provide an executive-level summary of portfolio size, loan exposure, default risk, and primary segment differences.

#### Key components

- Five KPI cards for Total Loan Amount, Average Loan Amount, Default Rate, Average Interest Rate, and Defaulted Loans.
- Default Rate Trend line chart for time-based risk monitoring.
- Loan Exposure by Purpose comparison.
- Average Income by Employment analysis.
- Loan Exposure by Age Group.
- Bookmark-based filter panel with dropdown slicers.
- Dynamic Key Insights area.

#### Recommended visual logic

Use cyan for portfolio exposure, purple/blue for supporting applicant metrics, orange for warnings, and red only for default/risk values. Keep labels white or light gray on the dark background. Avoid heavy borders and excessive neon effects.

#### Main interaction features

The page uses a hidden filter panel controlled by bookmarks. The  $\equiv$  FILTERS button opens the panel and the close button hides it. Bookmark Data is kept off so opening/closing the panel does not reset slicer selections. Page navigation is shared across the report.

#### Representative measures

Total Loans = COUNTROWS('Loan+Dataset+Link')

Total Loan Amount = SUM('Loan+Dataset+Link'[LoanAmount])

Average Loan Amount = AVERAGE('Loan+Dataset+Link'[LoanAmount])

Defaulted Loans = CALCULATE([Total Loans], 'Loan+Dataset+Link'[Default] = TRUE())

Default Rate = DIVIDE([Defaulted Loans], [Total Loans], 0)

Average Interest Rate = AVERAGE('Loan+Dataset+Link'[InterestRate]) / 100

## 4. Dashboard Page 2 - Applicant Profile & Financial Behavior

Purpose: understand applicant composition and financial behavior without duplicating the executive summary from Page 1.

### Recommended analytical visuals

- Applicant summary KPIs: Total Applicants, Average Income, Average Loan Amount, Average Credit Score, and High Credit Applicants.
- Matrix-style conditional-formatting view to compare loan exposure across Age Group and Employment Type.
- Decomposition Tree for interactive loan-exposure breakdown across Education, EmploymentType, Age Group, Credit score Bins, LoanPurpose, and MaritalStatus.
- Treemap for applicant education composition.
- Average Income by Employment as a supporting comparison visual.

### Why the page uses different visual types

The page intentionally avoids a grid of repetitive bar charts. The matrix supports segment comparison, the decomposition tree supports interactive diagnostic exploration, and the treemap communicates part-to-whole composition. Together these visuals demonstrate broader Power BI capability.

### Important design rule

Avoid putting a unique applicant/LoanID field into a scatter visual when a compact segment view is required. A high-cardinality Details/Values field can produce thousands of overlapping points. For this dataset, segmented matrix and decomposition visuals provide clearer portfolio insight.

### Suggested KPI measures

```
Total Applicants = COUNTROWS('Loan+Dataset+Link')
```

```
Average Income = AVERAGE('Loan+Dataset+Link'[Income])
```

```
Average Credit Score = AVERAGE('Loan+Dataset+Link'[CreditScore])
```

```
High Credit Applicants = CALCULATE([Total Applicants], 'Loan+Dataset+Link'[Credit score Bins] = "High")
```

## 5. Dashboard Page 3 - Credit Risk Intelligence

Purpose: focus the viewer on default risk, high-risk exposure, and factors associated with default.

### Risk KPIs

- Low Credit Exposure - ₹14.9bn.
- Low Credit Exposure % - 45.7%.
- Overall Default Rate - 11.61%.
- Low Credit Default Rate - 12.58%.
- Defaulted Loans - 29.65K.

### Key Influencers

The native Power BI Key Influencers visual is used to explore factors associated with Default = True. The configured explanatory fields include Credit score Bins, Income Bracket, EmploymentType, DTIRatio, InterestRate, Age Group, Education, MaritalStatus, LoanPurpose, LoanTerm, HasMortgage, and HasDependents.

Current analytical findings shown by Key Influencers include: Low Income = 2.22× higher likelihood of default in the current analysis, and Teen = 2.17× higher likelihood. These are associations surfaced by the model, not causal claims.

### Risk measures

A low-credit exposure measure is defined as the total loan amount for Low and Very Low credit-score bins. The corresponding exposure percentage is the low-credit exposure divided by total loan amount. The low-credit default rate is calculated by applying the same segment filter to the overall default-rate measure.

### Representative DAX

```
Low Credit Exposure = CALCULATE( [Total Loan Amount], 'Loan+Dataset+Link'[Credit score Bins] IN {"Low", "Very Low"} )
```

```
Low Credit Exposure % = DIVIDE([Low Credit Exposure], [Total Loan Amount], 0)
```

```
Low Credit Default Rate = CALCULATE( [Default Rate], 'Loan+Dataset+Link'[Credit score Bins] IN {"Low", "Very Low"} )
```

### Key Risk Insights panel

The bottom-right insight panel summarizes the most decision-relevant values on Page 3: overall default rate, low-credit exposure, low-credit default rate, and the strongest Key Influencers. Insights should remain concise and should clearly distinguish descriptive metrics from model associations.

## 6. Visual Design System

The same design system is carried across all three pages to create a consistent portfolio identity.

### Color palette

| Usage             | HEX     |
|-------------------|---------|
| Main background   | #071225 |
| Visual background | #0E1A33 |
| KPI cards         | #111C35 |
| Border            | #263B63 |
| Primary cyan      | #00D9C0 |
| Purple            | #7C3AED |
| Blue              | #2196F3 |
| Warning orange    | #FFAA00 |
| Risk red          | #FF3B4E |
| Primary text      | #FFFFFF |
| Secondary text    | #CBD5E1 |
| Muted text        | #94A3B8 |

### Typography

Recommended typography is Segoe UI / Aptos style: 22-24 px page titles, 14-16 px visual titles, 24-30 px KPI values, 10-12 px labels and 9-11 px axis text. The visual hierarchy is intentionally simple so the data remains the primary focus.

### Interaction design

- Bookmark-based filter panel.
- Consistent dropdown slicers.
- Page navigation across the three report pages.
- Cross-filtering between visuals.
- Tooltips for additional context.
- Drill-through concept for segment-level risk analysis.

### Portfolio presentation principles

1. Keep decoration behind the data.
2. Use semantic colors: cyan for normal portfolio information, orange for attention, red for risk/default.
3. Avoid excessive chart density.
4. Use concise business titles instead of raw aggregation names.
5. Prefer insight-driven visuals over chart count.

## 7. Power BI Service & GitHub Deployment

The final report was published to Power BI Service under the report name Loan Portfolio & Default Analytics. The project PBIX is stored in a public GitHub repository for portfolio sharing.

### Power BI Service objects

The final deployment uses a dedicated semantic model and report separate from the older working Dashboard objects. The older model/report was kept as a temporary backup during the transition and validation process.

### Data refresh limitation

The upstream Power BI Dataflow contains features that cannot be refreshed in the current workspace without Premium capacity. Because of this limitation, Scheduled Refresh is not enabled for the portfolio deployment. A future production setup can remove the dependency by connecting the model directly to SQL Server through a supported gateway configuration or by using an appropriately licensed capacity.

### GitHub structure

The repository is organized as follows:

```
loan-portfolio-default-analytics/  
  Data/  
  README.md  
  Documentation/  
  project-documentation.pdf  
  PowerBI/  
  Loan Portfolio & Default Analytics.pbix  
  Screenshots/  
  page-1-overview.png  
  page-2-applicant-profile.png  
  page-3-credit-risk.png  
  README.md
```

### Recommended repository contents

The repository should include the final PBIX, clean page screenshots, this project documentation, and a README containing the project objective, dashboard preview, tools, features, and live Power BI report link. The raw dataset should only be uploaded when redistribution rights allow it.

## 8. Portfolio Takeaways

The completed project demonstrates a workflow beyond basic chart building: data preparation, reusable DAX measures, segmentation, interactive report design, business storytelling, Power BI Service publication, and GitHub-based project presentation.

The strongest portfolio talking points are:

- Building a consistent dark financial dashboard theme across multiple pages.
- Using DAX measures for reusable KPIs and segment-level risk metrics.
- Implementing bookmark-driven filter interactions.
- Using Key Influencers to move from descriptive reporting toward diagnostic analysis.
- Using Decomposition Tree to allow interactive breakdowns of loan exposure.
- Separating executive summary, applicant behavior, and risk intelligence into three distinct analytical questions.

### Current data-driven observations

- Overall default rate: 11.61%.
- Defaulted loans: 29.65K.
- Low + Very Low credit exposure: ₹14.9bn, or 45.7% of total loan exposure under the current model definition.
- Low-credit default rate: 12.58%, higher than the overall portfolio rate shown in the report.
- Key Influencers identifies Low Income (2.22x) and Teen (2.17x) as higher-likelihood segments in the current analysis.

### Next-step opportunities

For a future version, add a dedicated Date table, formalize sort-order columns for risk bands, create dynamic narrative measures for the insight panels, validate all time-intelligence measures against a proper calendar, and replace the Premium-dependent Dataflow refresh path with a sustainable refresh architecture if live updates are required.

**Project title: Loan Portfolio & Default Analytics**

**Primary tools: Power BI, DAX, Power Query, SQL Server, GitHub**

**Purpose: Portfolio-ready data analytics and credit-risk visualization project**